Week 8 — Drift and Observability Monitoring

Course: DS5619 Machine Learning Systems Operations
Student ID: `142301024`
Seed: `706208327` (salt: `week08`)

What Was Implemented

In this lab, I built an unsupervised drift monitoring pipeline in `src/drift_monitor.py` to detect distribution shifts between reference daylight CCTV camera data and live low-light camera data using the Population Stability Index (PSI).

The core functions implemented are:

1. **`extract_confidence_scores(camera_dir)`**:
   - Loops over all `.jpg` images in sorted alphabetical order from the camera directory.
   - Runs `mock_detector.detect()` on each image.
   - Extracts and returns a flat list of float confidence scores for all detected vehicles.
2. **`summarize_scores(scores)`**:
   - Calculates basic descriptive statistics: `count`, `mean`, sample standard deviation (`std` with Bessel's correction $N-1$), `min`, and `max`.
   - Returns 0.0 values gracefully if the score list is empty or has fewer than 2 elements.
3. **`compute_psi(reference_scores, live_scores, n_bins=10)`**:
   - Bins confidence scores into 10 equal-width bins across $[0.0, 1.0]$ (e.g. $[0.0, 0.1), [0.1, 0.2), ...$).
   - Calculates the proportion of samples falling into each bin for both reference and live distributions.
   - Applies a small regularization floor (`1e-4`) to every bin proportion so we never take $\ln(0)$ or divide by zero.
   - Sums the symmetric divergence formula:

$$\text{PSI} = \sum_{b=1}^{B} (P_{\text{live},b} - Q_{\text{ref},b}) \times \ln\left(\frac{P_{\text{live},b}}{Q_{\text{ref},b}}\right)$$

4. **`classify_drift(psi)`**:
   - Maps the computed PSI value into standard alert thresholds:
     - `psi < 0.10`: `"none"`
     - `0.10 <= psi < 0.25`: `"moderate"`
     - `psi >= 0.25`: `"significant"`

Results

Running the pipeline (`python src/run_pipeline.py`) compared daylight reference images against live low-light images and generated `drift_report.json`:

Metric	Reference Camera (`camera_A_daylight`)	Live Camera (`camera_B_lowlight`)
Detection Count	73	75
Mean Confidence	0.9698	0.9767
Standard Deviation	0.0524	0.0290
Score Range	[0.6240, 0.9800]	[0.7290, 0.9800]

- **Calculated PSI:** `0.1346`
- **Drift Classification:** `moderate` (falls in the range $0.10 \leq \mathrm{PSI} < 0.25$)

Key Takeaway:

Low-light camera conditions shifted the confidence distribution enough to cross the 0.10 threshold into **moderate drift**. While not yet a critical failure (significant drift ≥ 0.25), this alerts the engineering team that the model's environment has shifted and should be monitored for potential retraining or calibration.